



BAYESIAN DATA ANALYSIS

BREAST CANCER RISK PREDICTION USING BAYESIAN LOGISTIC REGRESSION

Group 2

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DATASET

DESCRIPTION

The dataset used in this study is the Wisconsin Diagnostic Breast Cancer (WDBC) dataset, originally obtained from the University of Wisconsin–Madison and publicly available through the UC Irvine Machine Learning Repository. A widely used version of this dataset is also available on Kaggle. The dataset contains features computed from digitized images of fine needle aspirates (FNA) of breast masses, with the objective of classifying tumors as malignant or benign.



FEATURES

Variable	Role	Type	Description	Null Values
id	Feature	Integer	Patient ID	No
diagnosis	Target	Categorical	M = malignant, B = benign	No
radius_mean	Feature	Numeric	Mean of distances from center to points on the tumor boundary	No
texture_mean	Feature	Numeric	Standard deviation of gray-scale values	No
perimeter_mean	Feature	Numeric	Mean perimeter of the tumor	No

FEATURES

Variable	Role	Type	Description	Null Values
area_mean	Feature	Numeric	Mean area of the tumor	No
smoothness_mean	Feature	Numeric	Mean local variation in radius lengths	No
compactness_mean	Feature	Numeric	Mean compactness (perimeter ² / area - 1.0)	No
concavity_mean	Feature	Numeric	Mean severity of concave portions of the contour	No

FEATURES

Variable	Role	Type	Description	Null Values
concave_mean	Feature	Numeric	Mean number of concave portions of the contour	No
symmetry_mean	Feature	Numeric	Mean of symmetry values of the tumor cell nuclei	No
fractal_dimension_mean	Feature	Numeric	Mean fractal dimension of the tumor cell nuclei boundary	No

METHODOLOGY

Bayesian Logistic Regression Model Specification

The study employs Bayesian Logistic Regression to model tumor malignancy (malignant = 1, benign = 0) as a function of standardized tumor features (radius mean, texture mean, and perimeter mean). Regression coefficients are assigned weakly informative Normal priors to incorporate parameter uncertainty and provide mild regularization. The binary outcome is modeled using a Bernoulli likelihood with a logit link function, allowing predictors to be interpreted on the log-odds scale.



METHODOLOGY

Likelihood

Bernoulli

$$y_i \sim \text{Bernoulli}(\pi_i)$$

with logit link function:

$$\text{logit}(\pi_i) = \log\left(\frac{\pi_i}{1-\pi_i}\right) = \beta_0 + \beta_1 x_{i1} + \dots + \beta_p x_{ip}$$

Prior

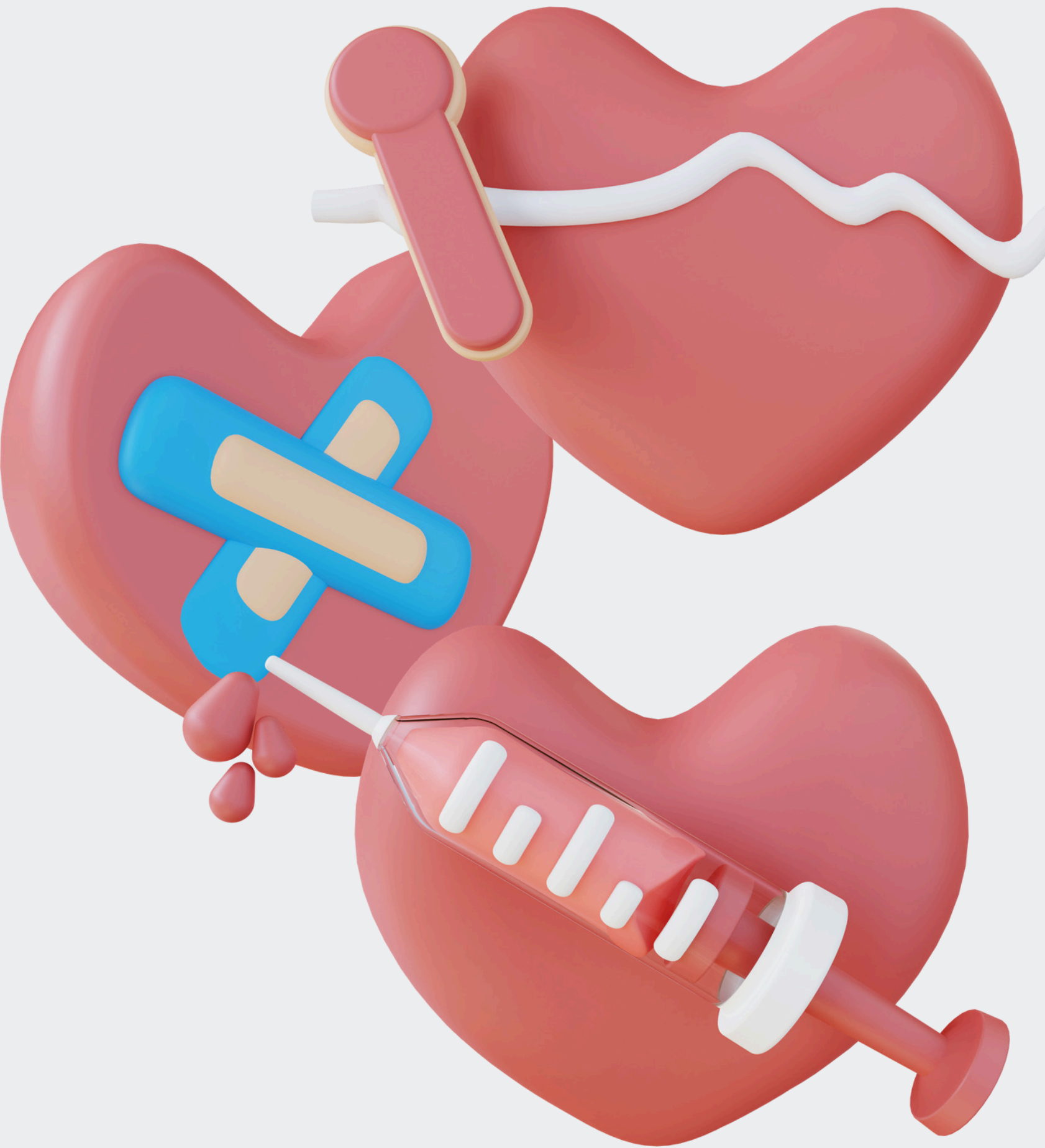
Normal Weakly Informative

$$\beta_i \sim N(0, \sigma^2)$$

Posterior Distribution

$$p(\beta | y) \propto p(y | \beta) p(\beta)$$

$$p(y | \beta) = \prod_{i=1}^n \pi_i^{y_i} (1 - \pi_i)^{1-y_i}$$



METHODOLOGY

Bayesian Inference via MCMC Sampling

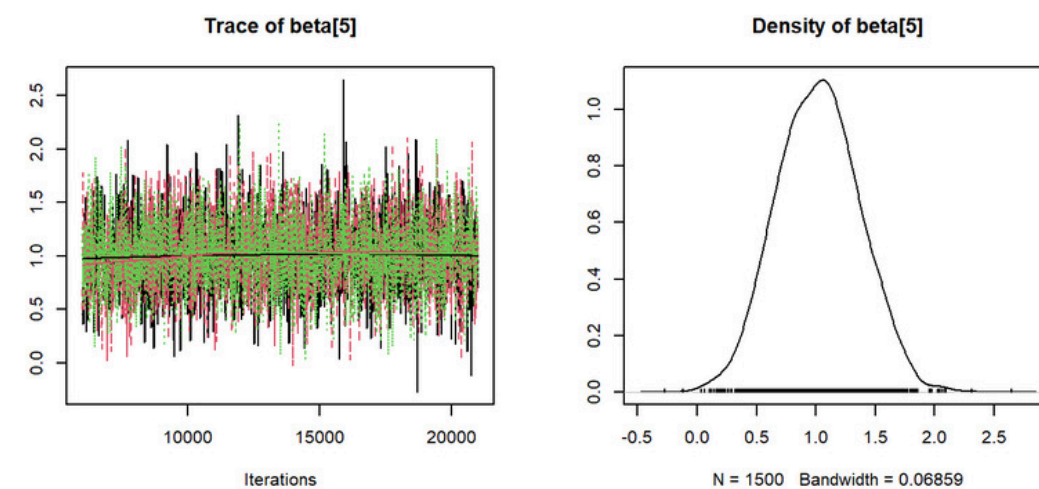
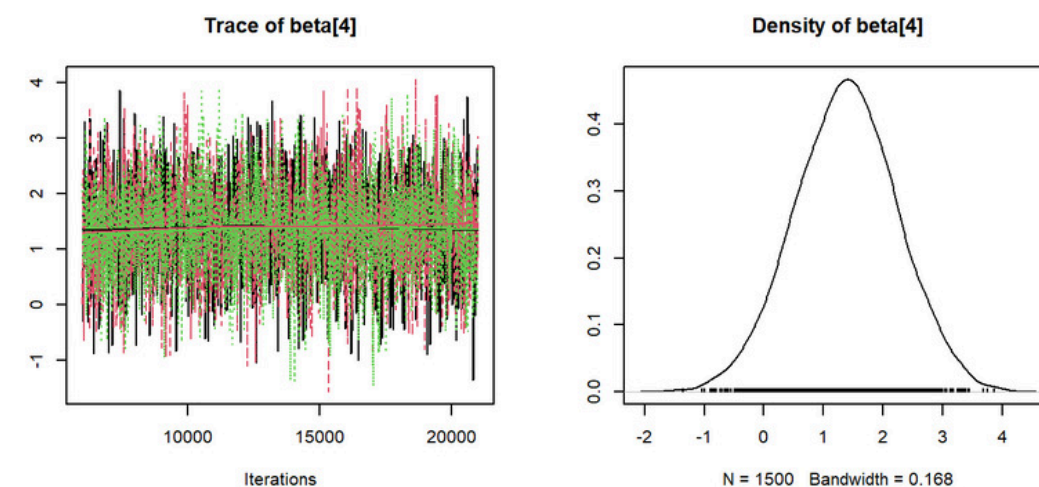
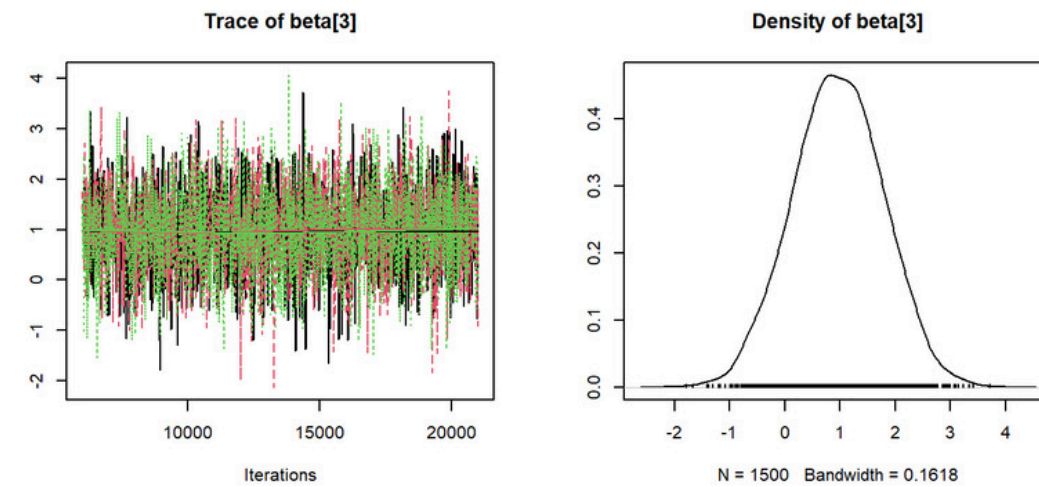
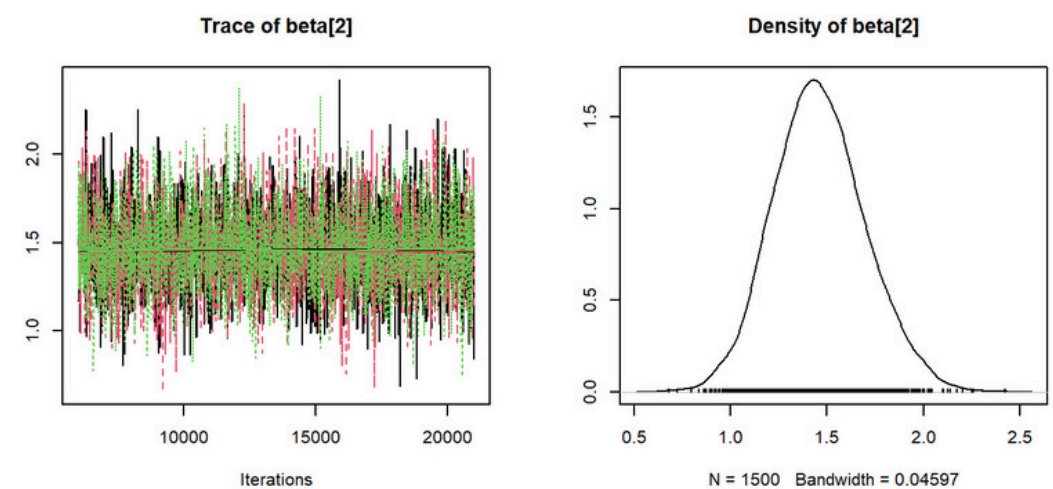
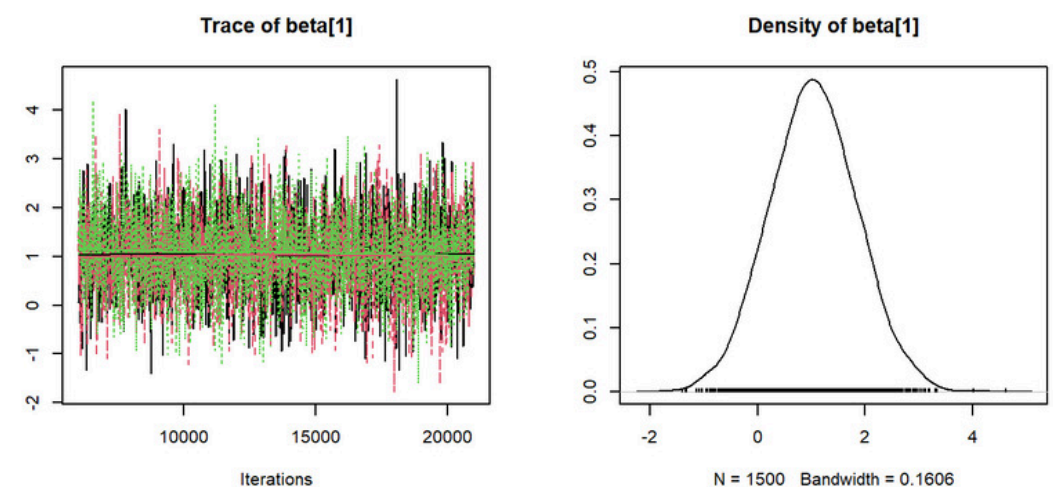
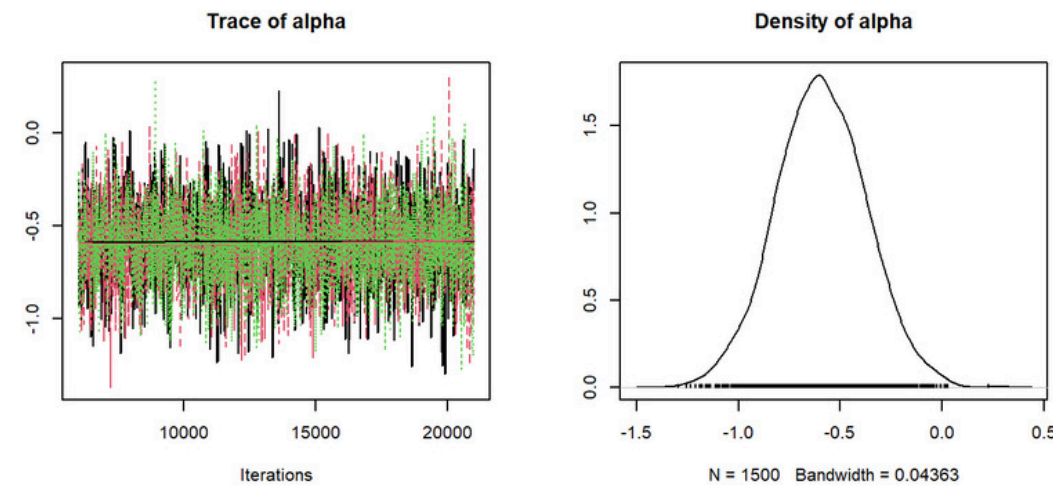
Posterior distributions of model parameters are estimated using Markov Chain Monte Carlo (MCMC) methods, specifically Hamiltonian Monte Carlo with the No-U-Turn Sampler (NUTS). Posterior summaries (means, credible intervals) are derived from the sampled distributions, and predictor effects are assessed based on whether the credible intervals (25%-97.5% posterior quantiles) exclude zero, providing a probabilistic measure of uncertainty in parameter estimates.

MODEL ESTIMATION RESULT

TRACE & DENSITY PLOT

The trace plots exhibit stable oscillations around constant mean levels across iterations, with no evident trends or long term drifts. This pattern indicates good mixing and suggests that the Markov chains have reached stationarity. The corresponding posterior density plots are smooth and unimodal, implying that the posterior distributions are well identified and that parameter estimates are stable.

Samples used: 15000
Burn-in period: 5000
Thinning: 10

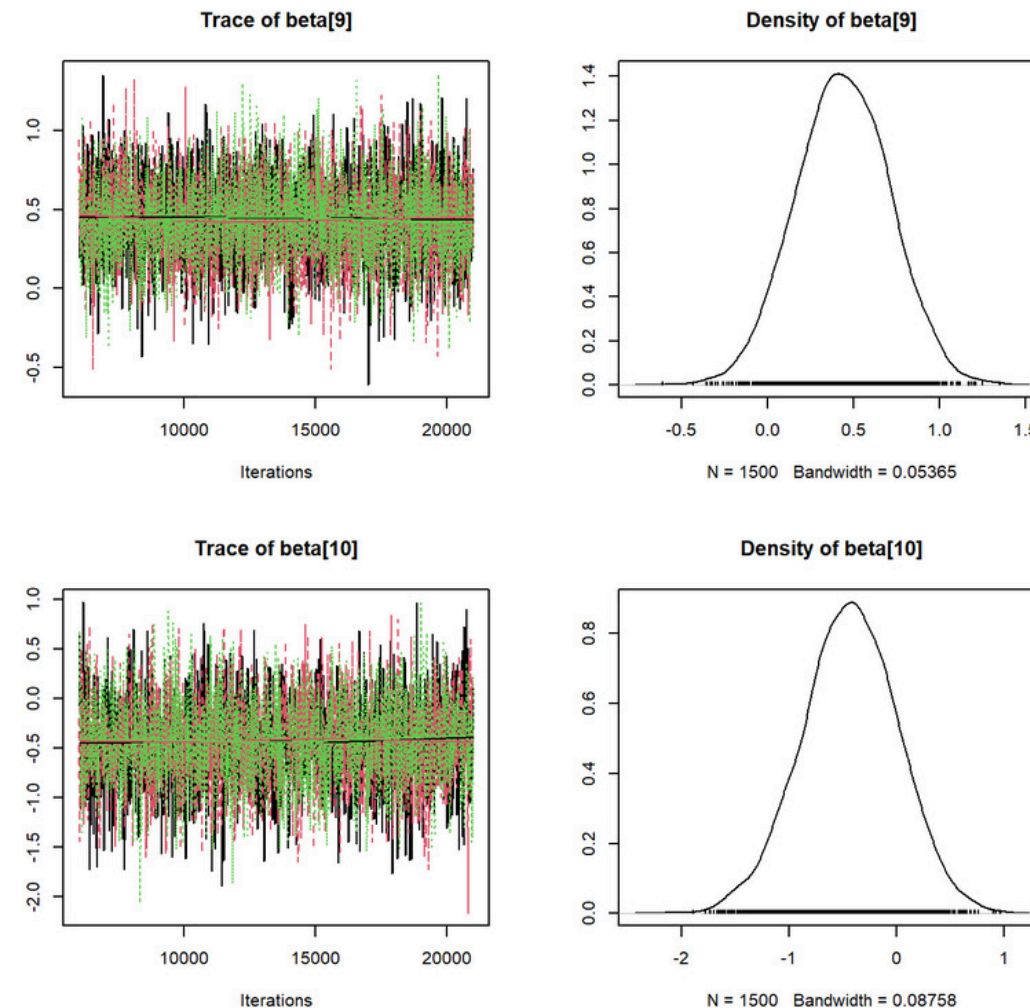
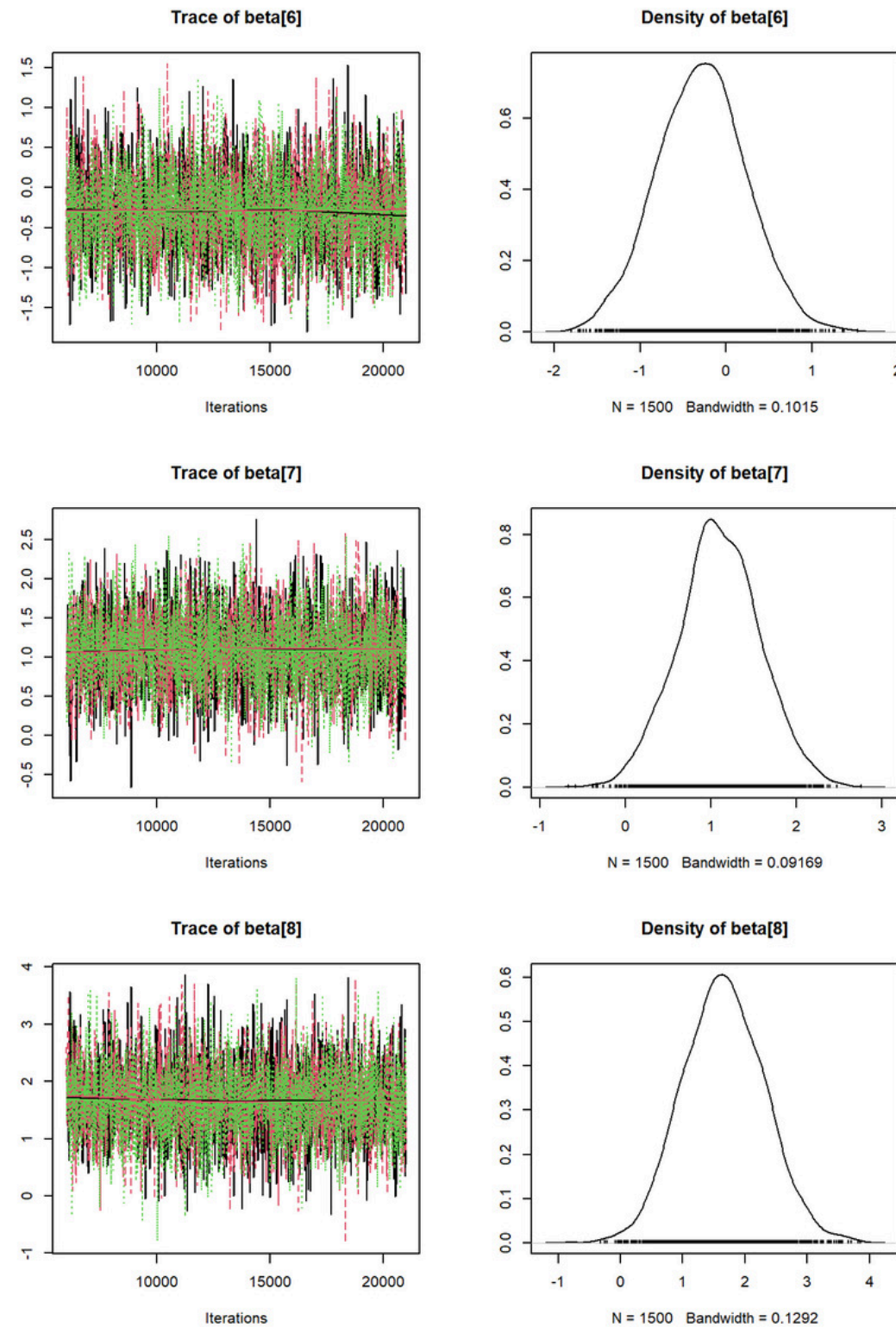


MODEL ESTIMATION RESULT

TRACE & DENSITY PLOT

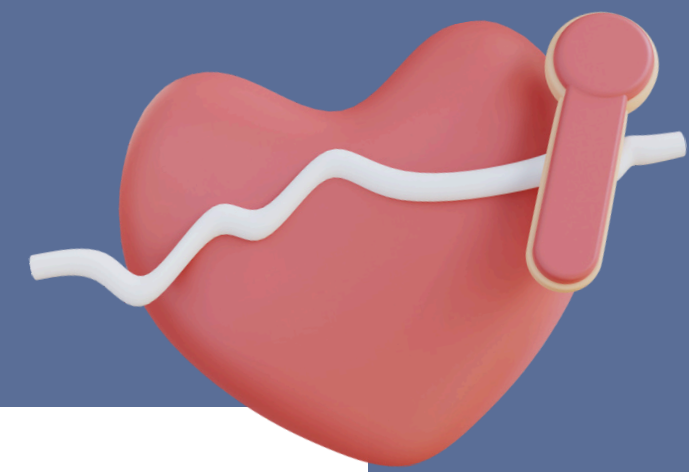
The trace plots exhibit stable oscillations around constant mean levels across iterations, with no evident trends or long term drifts. This pattern indicates good mixing and suggests that the Markov chains have reached stationarity. The corresponding posterior density plots are smooth and unimodal, implying that the posterior distributions are well identified and that parameter estimates are stable.

Samples used: 15000
Burn-in period: 5000
Thinning: 10





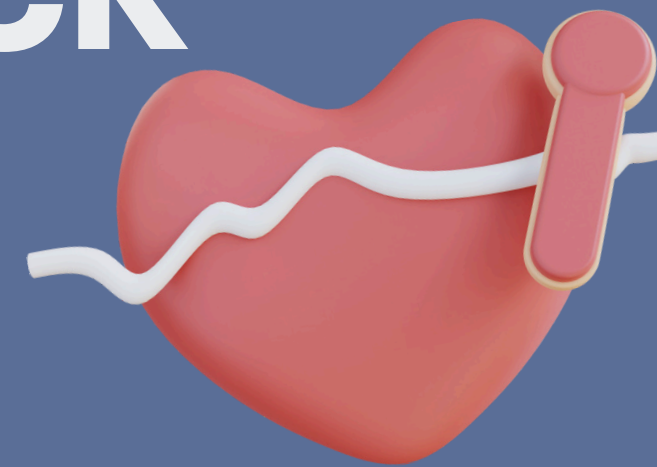
QUANTILES



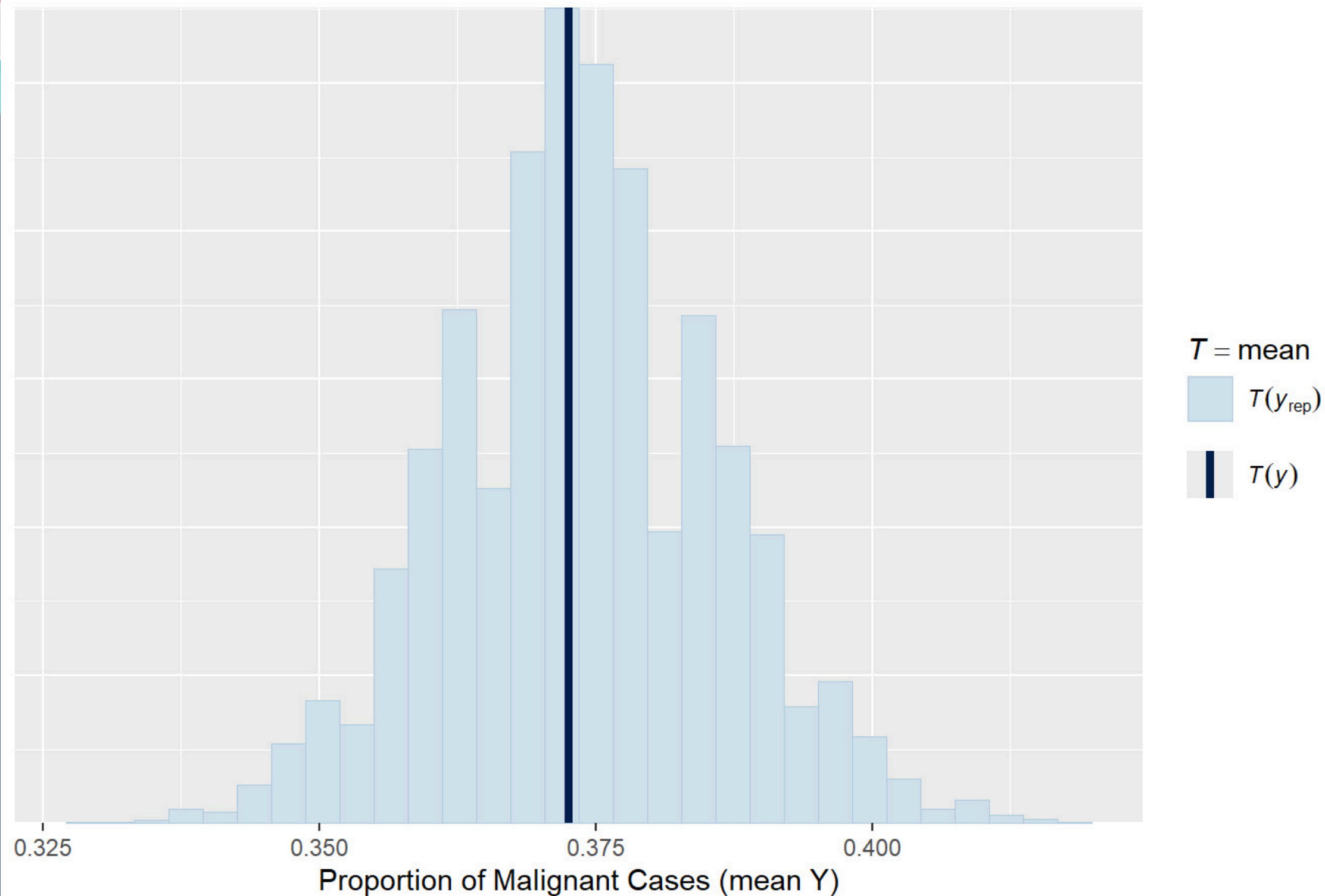
		25%	75%	97.5%
Intercept	alpha	-0.7397173	-0.44307972	-0.1567619
Radius_mean	beta[1]	0.4760613	1.58105402	2.6186403
Texture_mean	beta[2]	1.2986154	1.61250511	1.9366818
Perimeter_mean	beta[3]	0.3951012	1.53170099	2.5734221
Area_mean	beta[4]	0.8128375	1.95407490	3.0501690
Smoothness_mean	beta[5]	0.7879905	1.24878317	1.6979727
Compactness_mean	beta[6]	-0.6427112	0.06856243	0.7400516
Concavity_mean	beta[7]	0.7913220	1.43029290	2.0461699
Concave.points_mean	beta[8]	1.2074804	2.10352915	2.9826026
Symmetry_mean	beta[9]	0.2497773	0.61780713	0.9739884
Fractal_dimension_mean	beta[10]	-0.7310549	-0.12865603	0.4191266



POSTERIOR PREDICTIVE CHECK



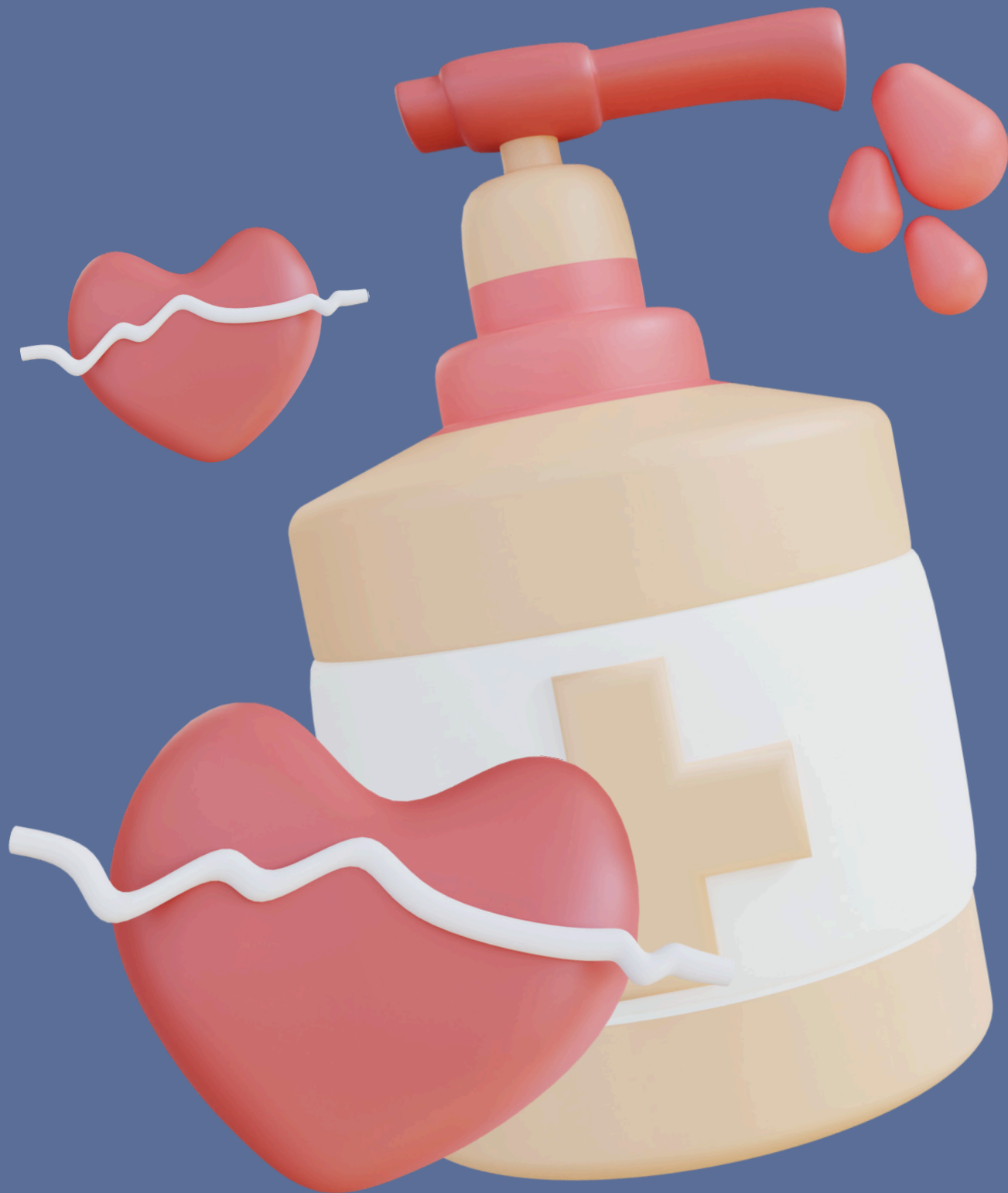
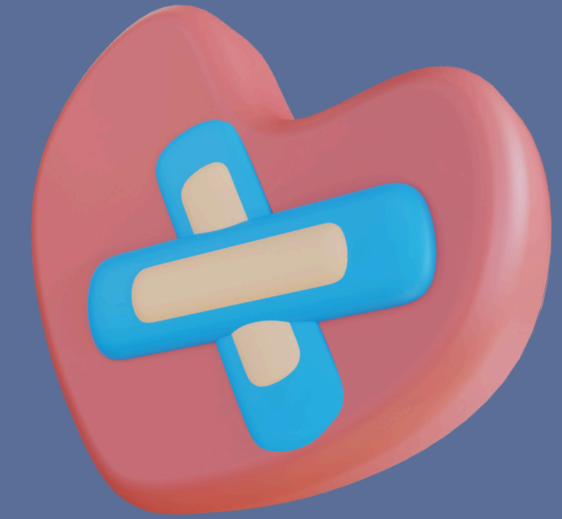
PPC: Distribution of Predicted Malignancy Rates



The PPC chart compares the mean from raw data with the mean from data produced by the model.

From the results, the data produced by the model fits the raw data well.

ACCURACY **94.55%**

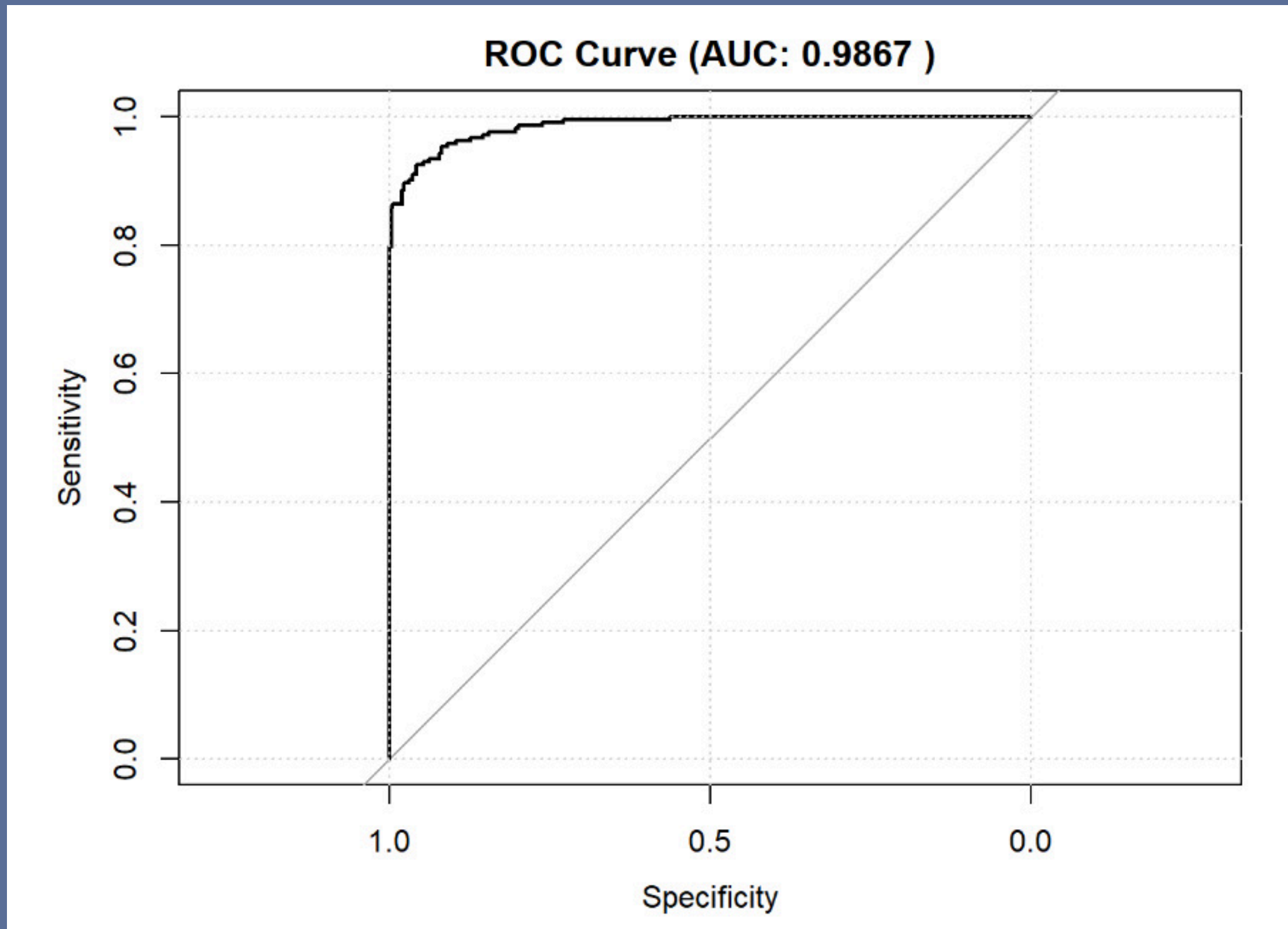


CONFUSION MATRIX

		Predicted		
Actual	0	1		
	0	348	9	
	1	22	190	

Obtained by converting probability prediction into binary outcomes (>0.5 to malignant, <= 0.5 to benign)

ROC CURVE AND AUC

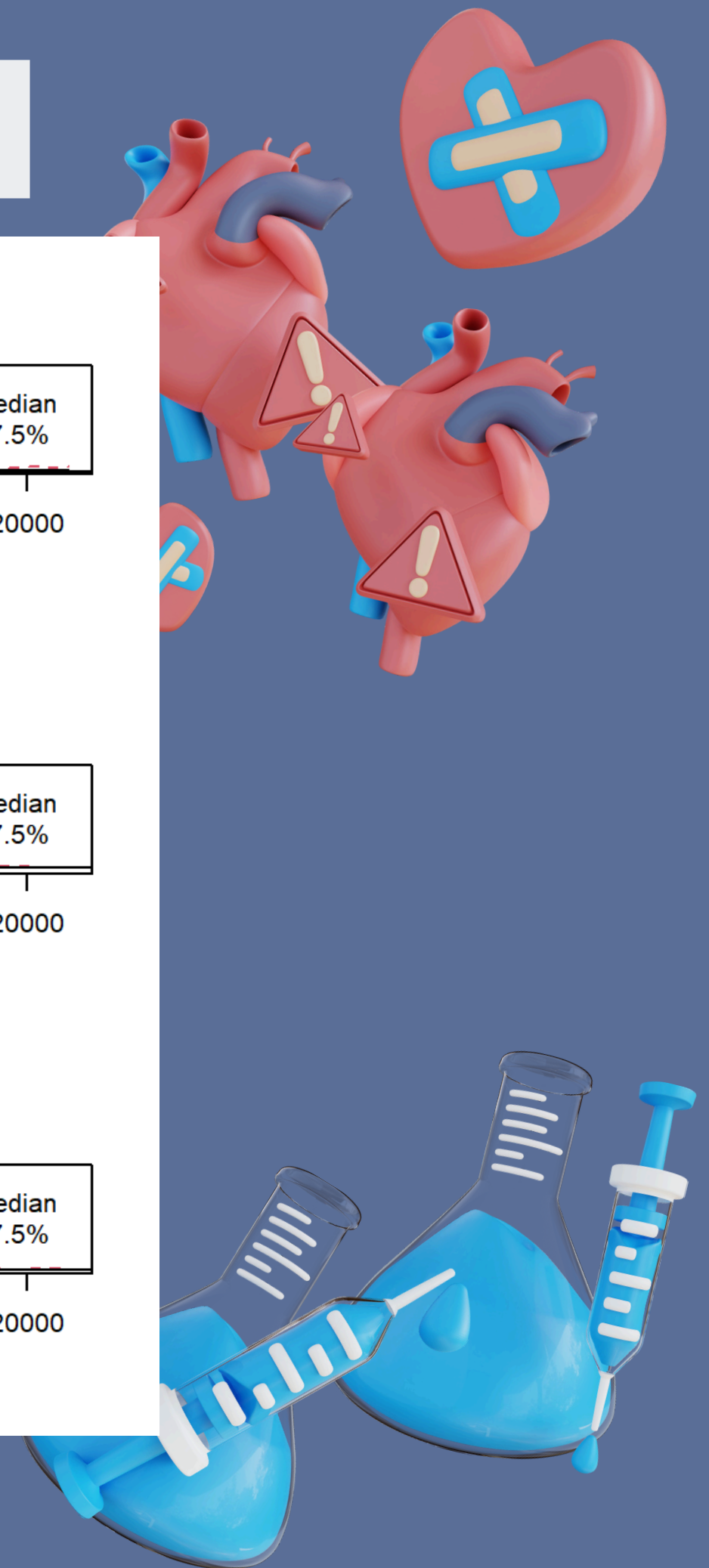
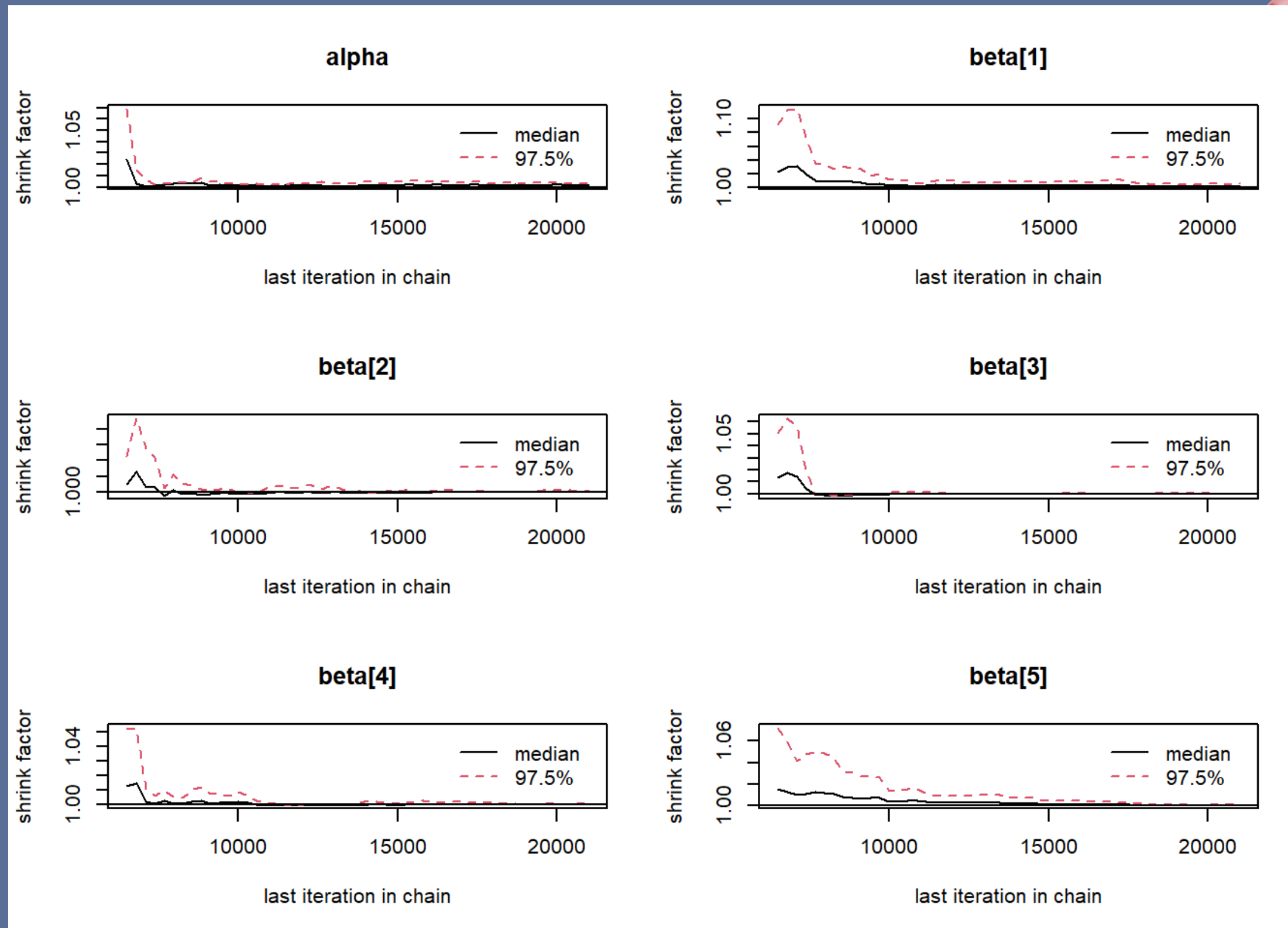


Sensitivity (Recall): 89.6%
 $TP / (TP + FN)$

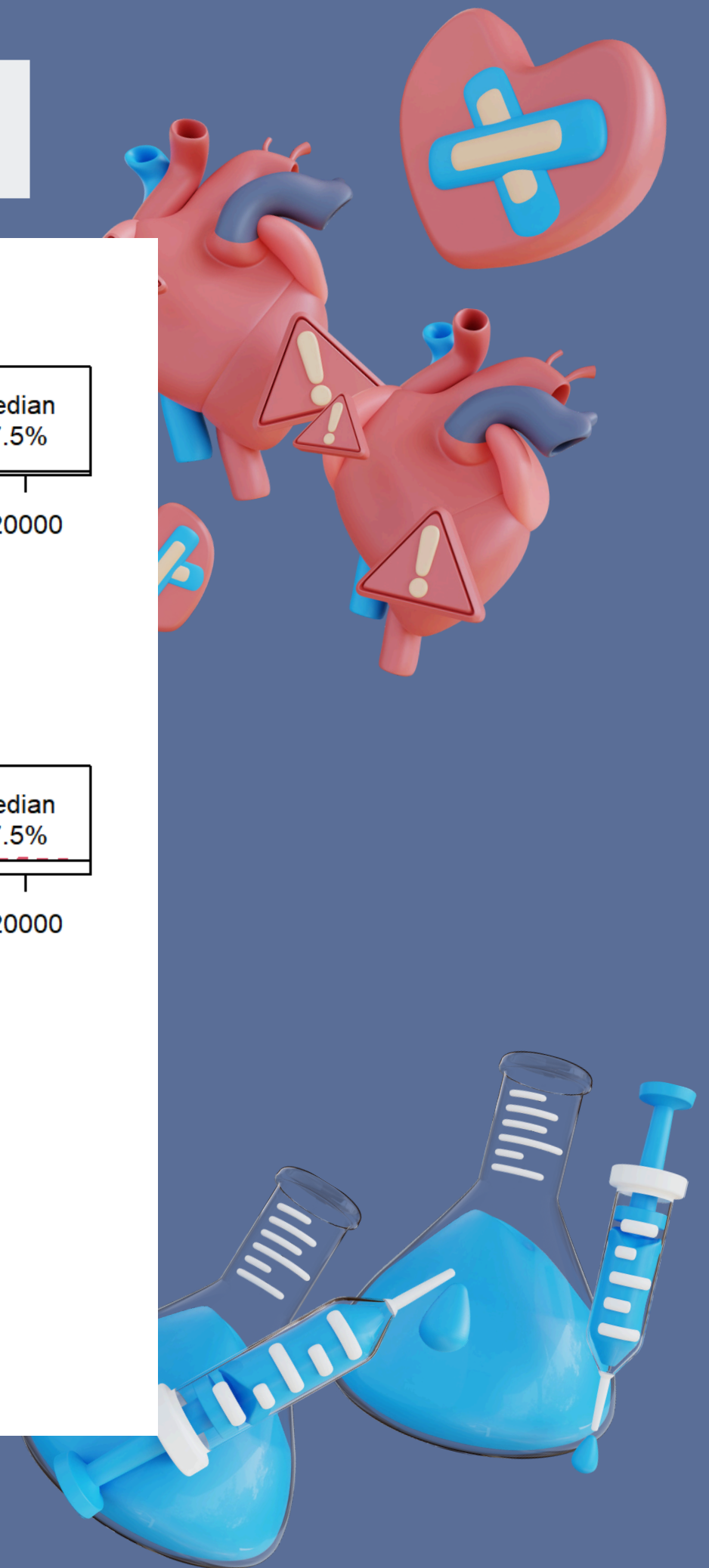
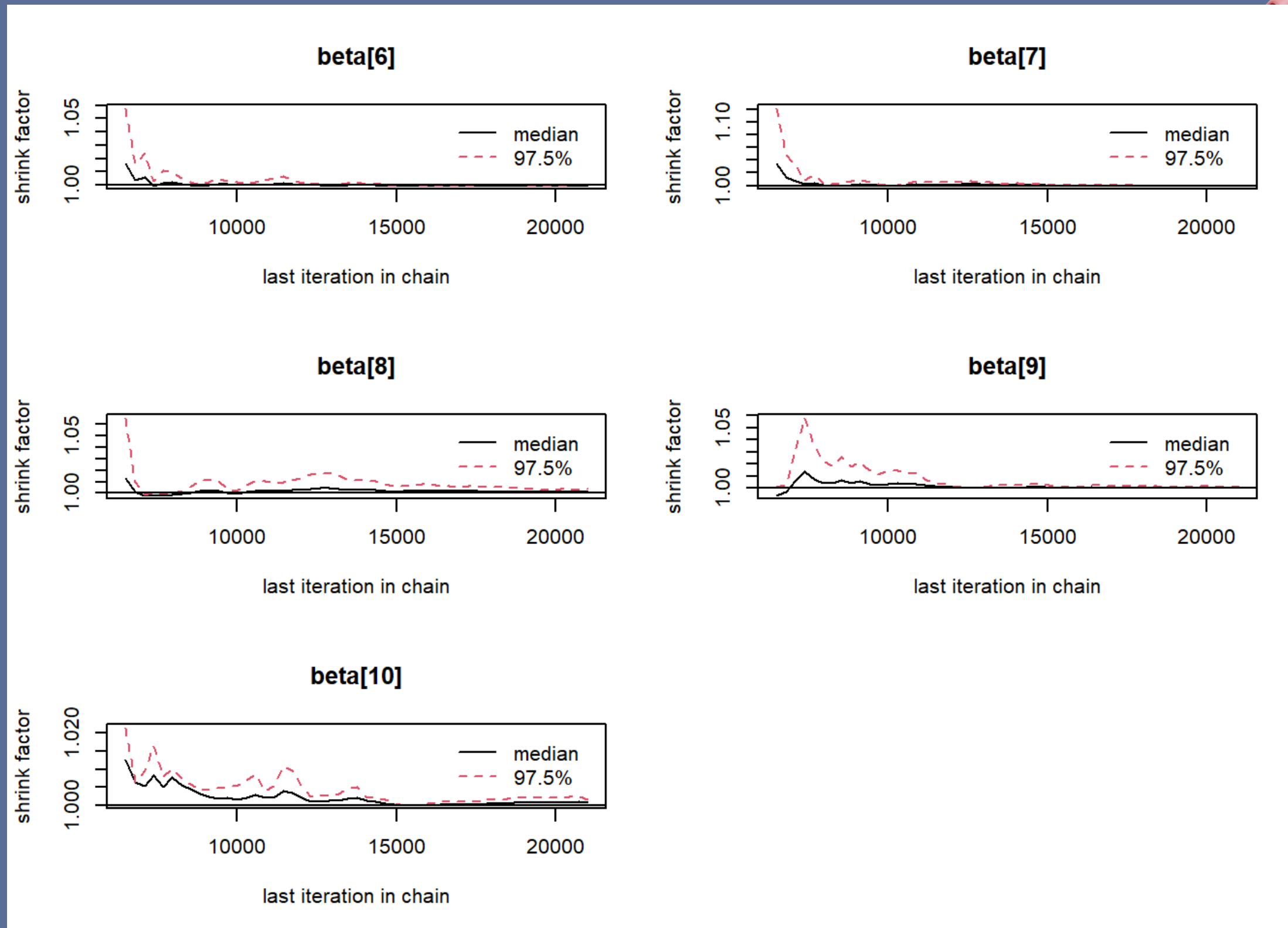
Specificity: 97.5%
 $TN / (TN + FP)$



GELMAN-RUBIN



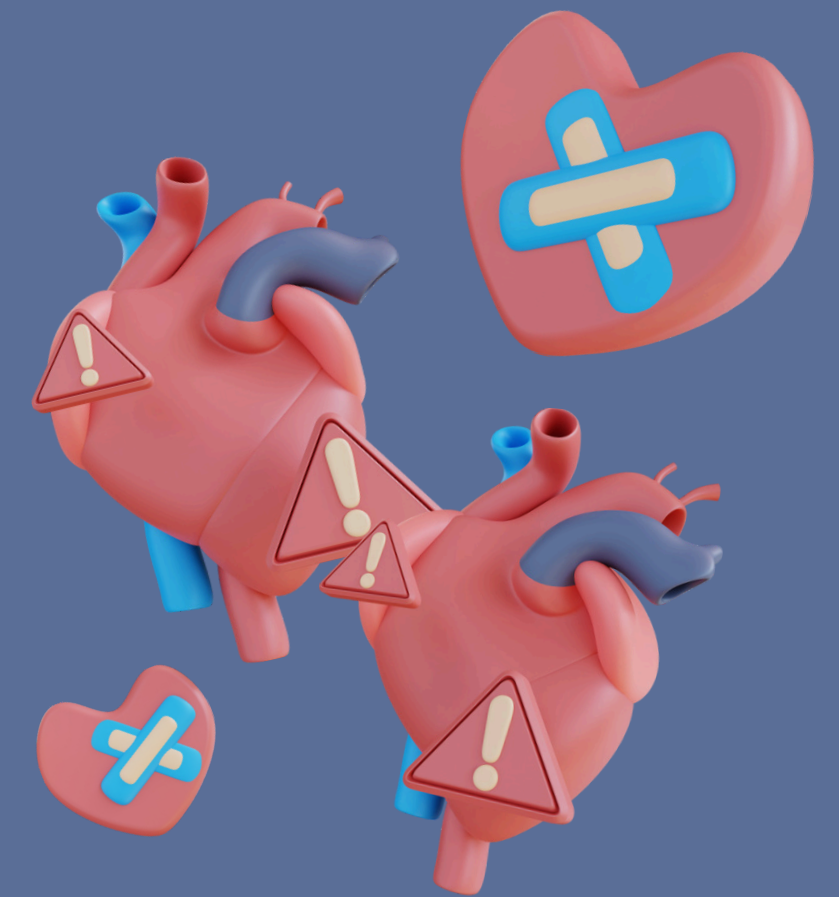
GELMAN-RUBIN



GELMAN-RUBIN

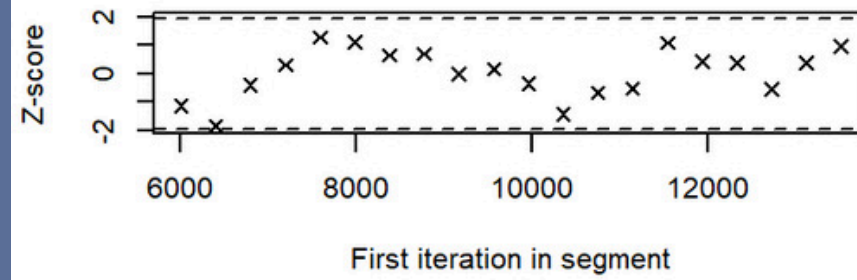
```
## Potential scale reduction factors:
##
##      Point est. Upper C.I.
## alpha          1          1
## beta[1]         1          1
## beta[2]         1          1
## beta[3]         1          1
## beta[4]         1          1
## beta[5]         1          1
## beta[6]         1          1
## beta[7]         1          1
## beta[8]         1          1
## beta[9]         1          1
## beta[10]        1          1
##
## Multivariate psrf
##
## 1
```

The Gelman-Rubin diagnostics results indicates that the MCMC simulations have reached optimal convergence to the target distribution

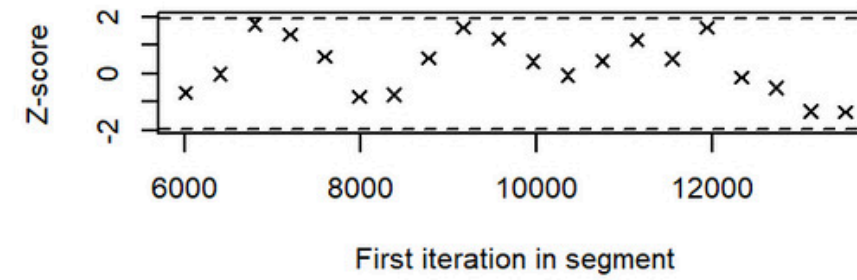


GEWEKE DIAGNOSTICS

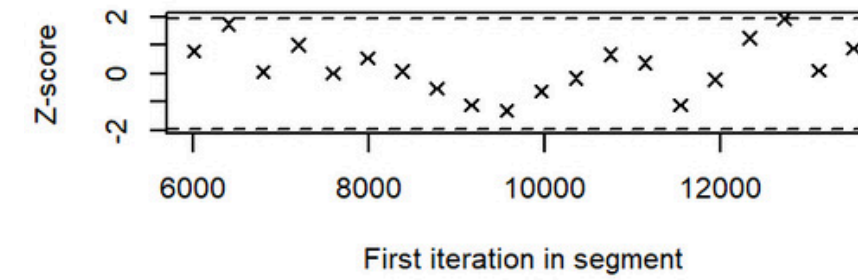
alpha (chain1)



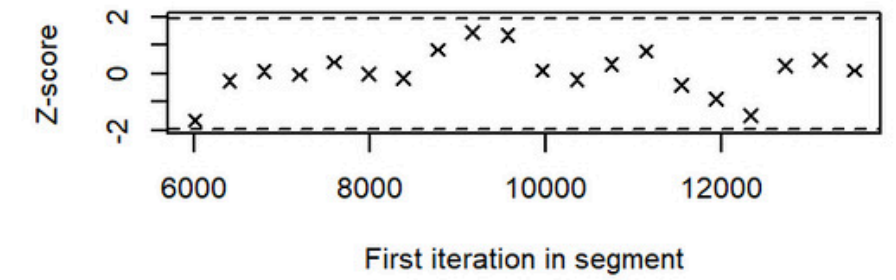
beta[1] (chain1)



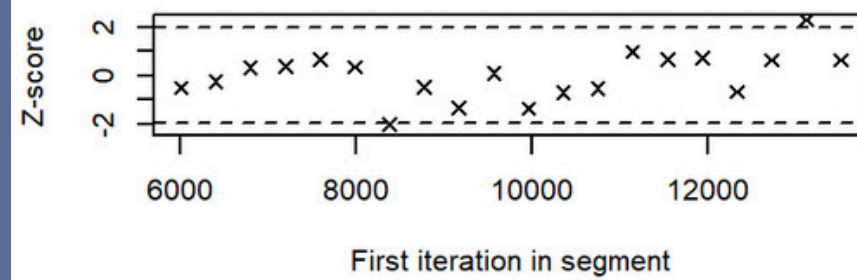
beta[6] (chain1)



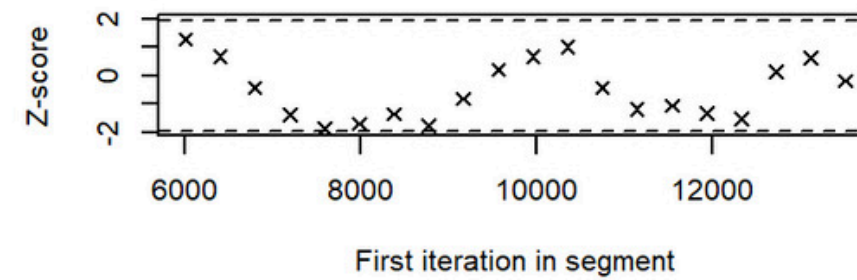
beta[7] (chain1)



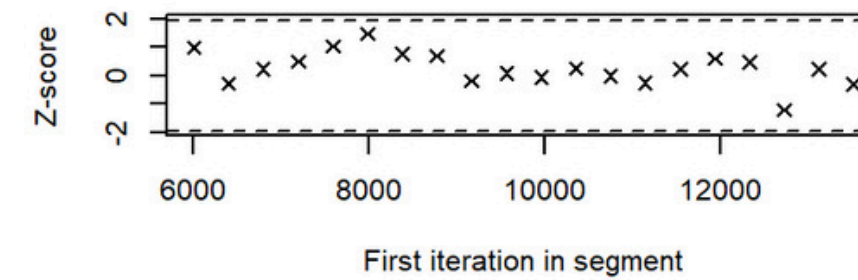
beta[2] (chain1)



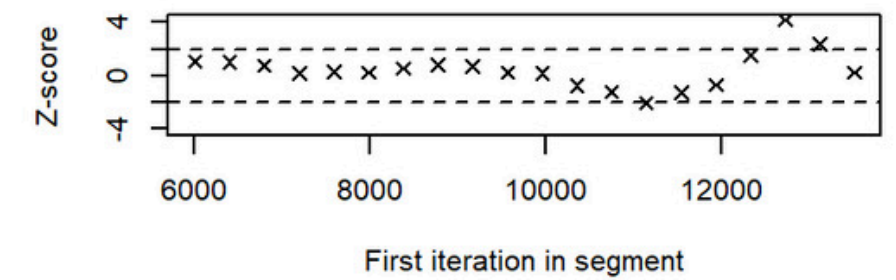
beta[3] (chain1)



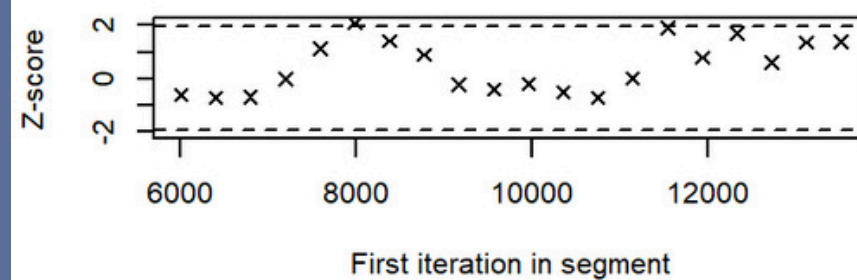
beta[8] (chain1)



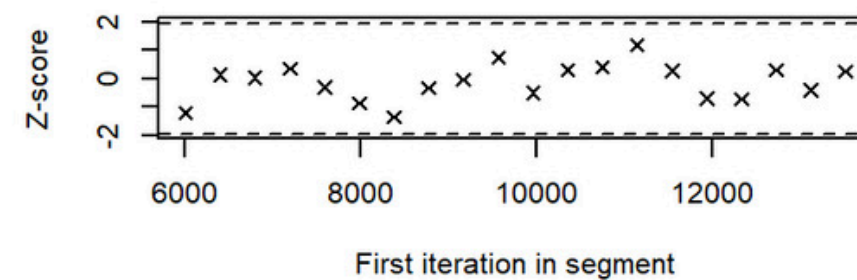
beta[9] (chain1)



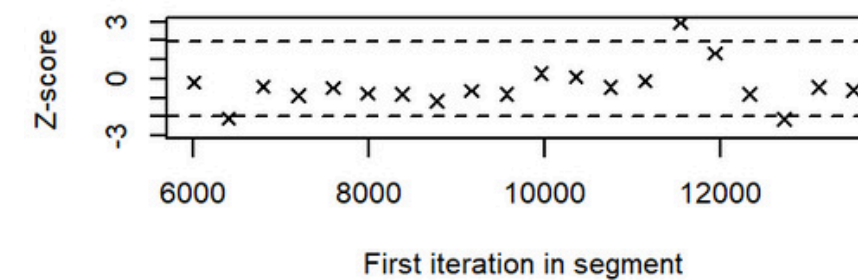
beta[4] (chain1)



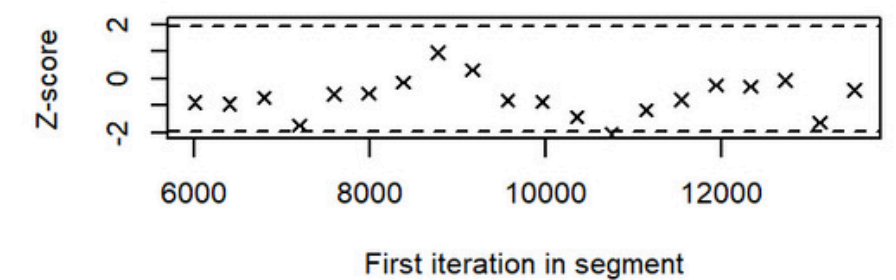
beta[5] (chain1)



beta[10] (chain1)



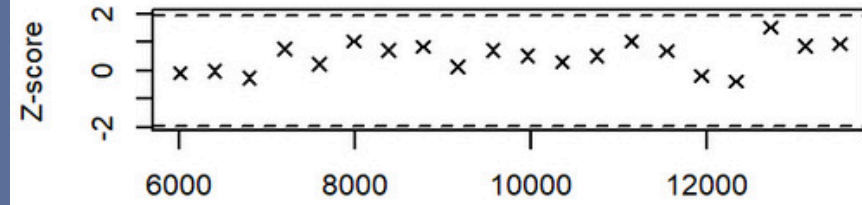
alpha (chain2)



```
## [[1]]
##
## Fraction in 1st window = 0.1
## Fraction in 2nd window = 0.5
##
##      alpha  beta[1]  beta[2]  beta[3]  beta[4]  beta[5]  beta[6]  beta[7]
## -1.1301 -0.6623 -0.4987  1.2894 -0.6044 -1.1967  0.8034 -1.6422
##      beta[8] beta[9] beta[10]
##  0.9841  1.0704 -0.1830
```

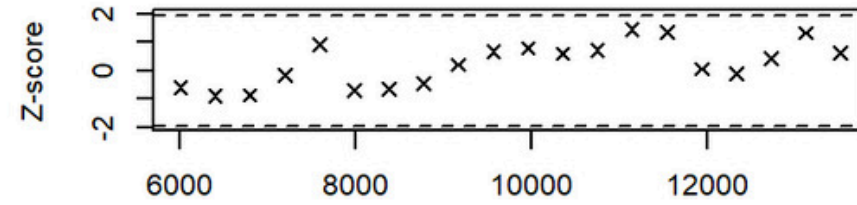

GEWEKE DIAGNOSTICS

beta[1] (chain2)



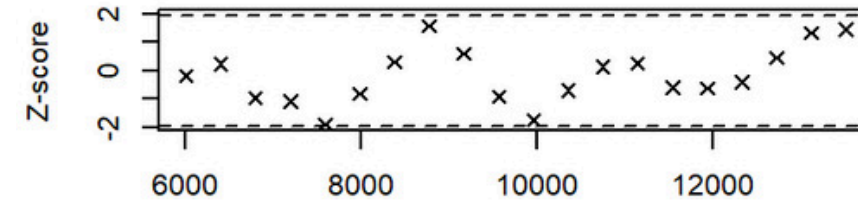
First iteration in segment

beta[2] (chain2)



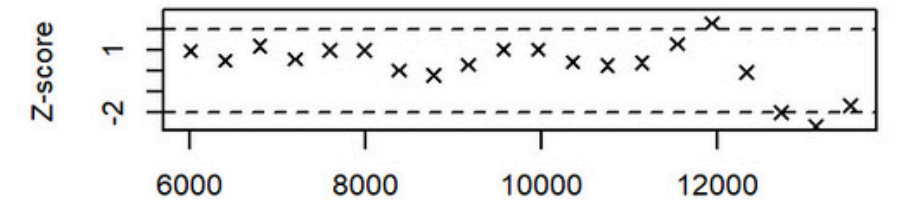
First iteration in segment

beta[7] (chain2)



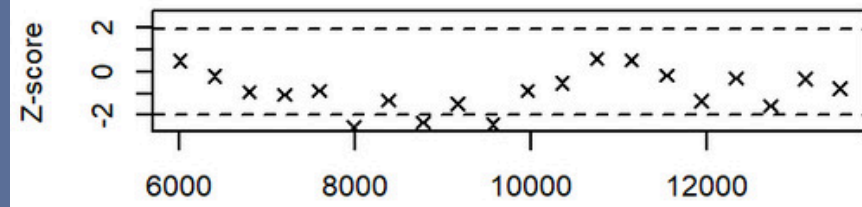
First iteration in segment

beta[8] (chain2)



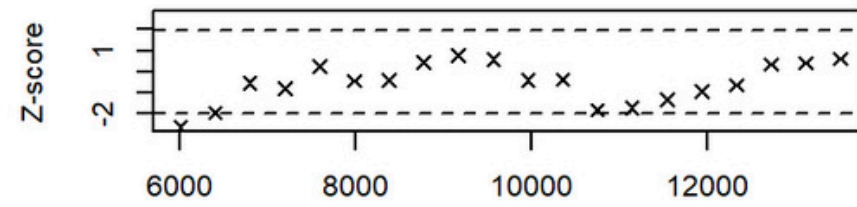
First iteration in segment

beta[3] (chain2)



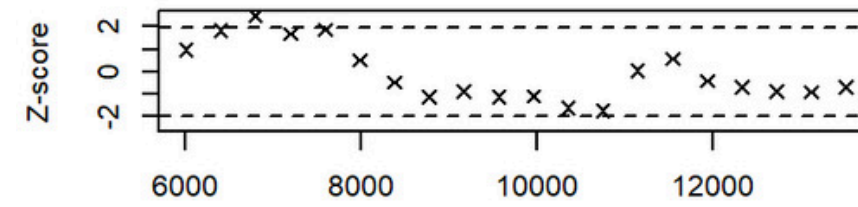
First iteration in segment

beta[4] (chain2)



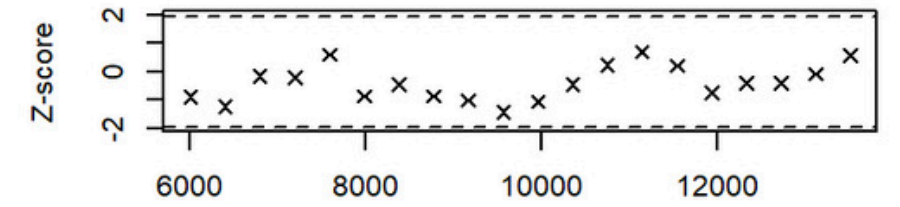
First iteration in segment

beta[9] (chain2)



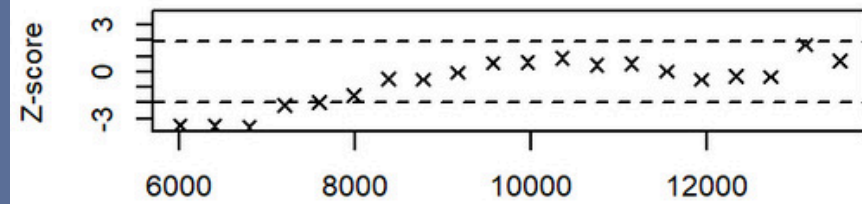
First iteration in segment

beta[10] (chain2)



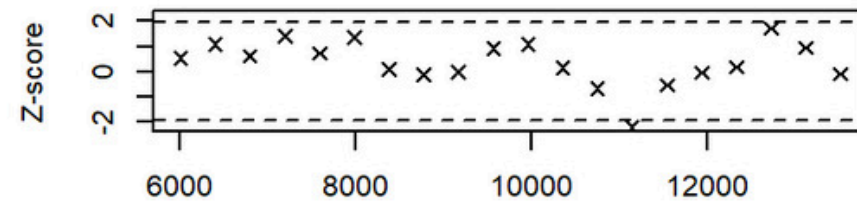
First iteration in segment

beta[5] (chain2)



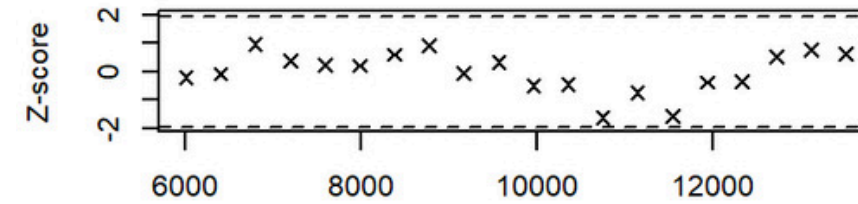
First iteration in segment

beta[6] (chain2)



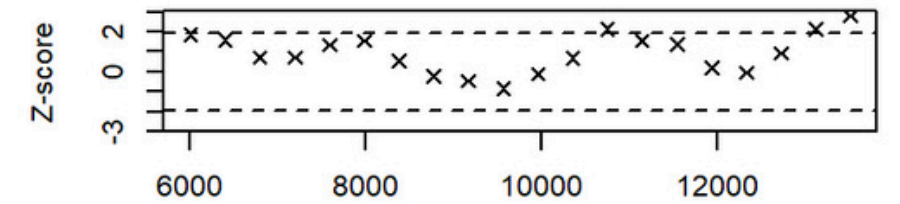
First iteration in segment

alpha (chain3)



First iteration in segment

beta[1] (chain3)

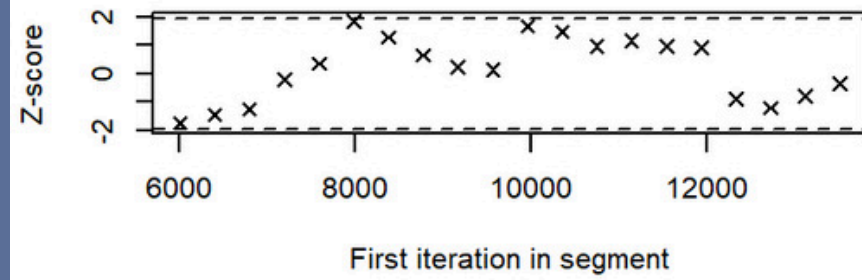


First iteration in segment

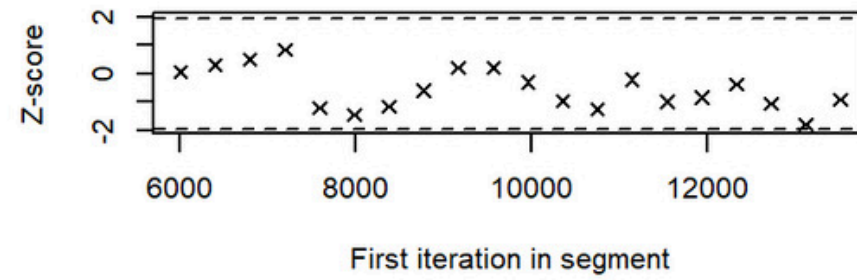
```
## [[2]]
##
## Fraction in 1st window = 0.1
## Fraction in 2nd window = 0.5
##
##      alpha  beta[1]  beta[2]  beta[3]  beta[4]  beta[5]  beta[6]  beta[7]
## -0.87773 -0.07689 -0.59272  0.50984 -2.63821 -3.43193  0.53729 -0.18562
##  beta[8]  beta[9] beta[10]
##  0.93442  0.98280 -0.88757
```

GEWEKE DIAGNOSTICS

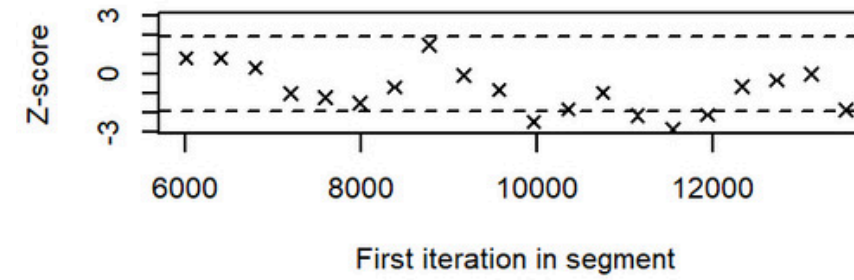
beta[2] (chain3)



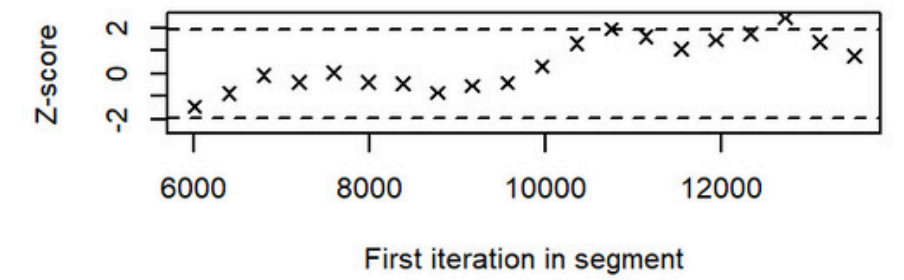
beta[3] (chain3)



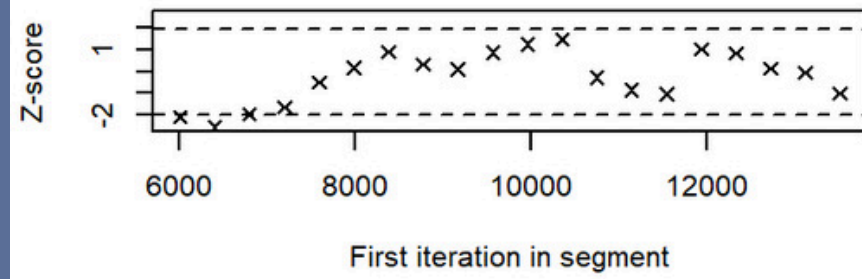
beta[8] (chain3)



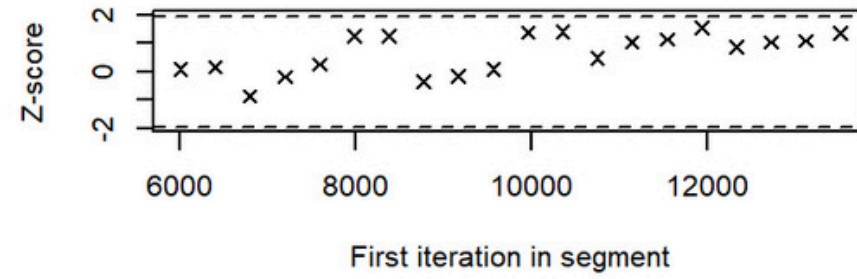
beta[9] (chain3)



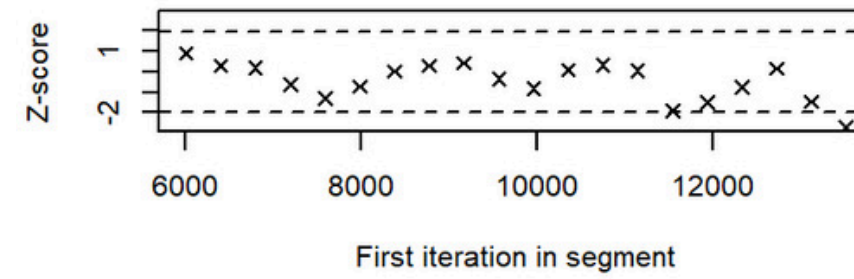
beta[4] (chain3)



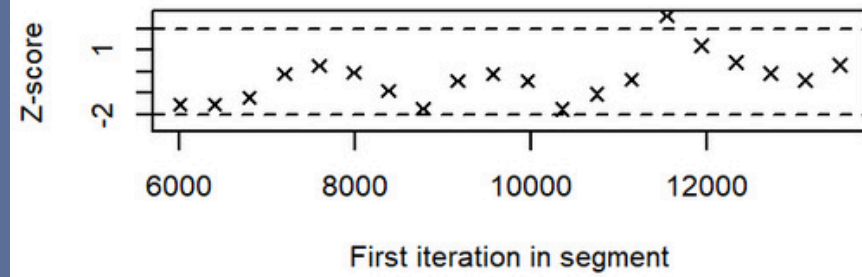
beta[5] (chain3)



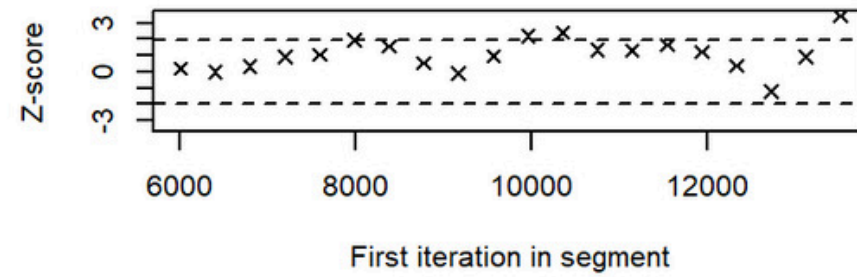
beta[10] (chain3)



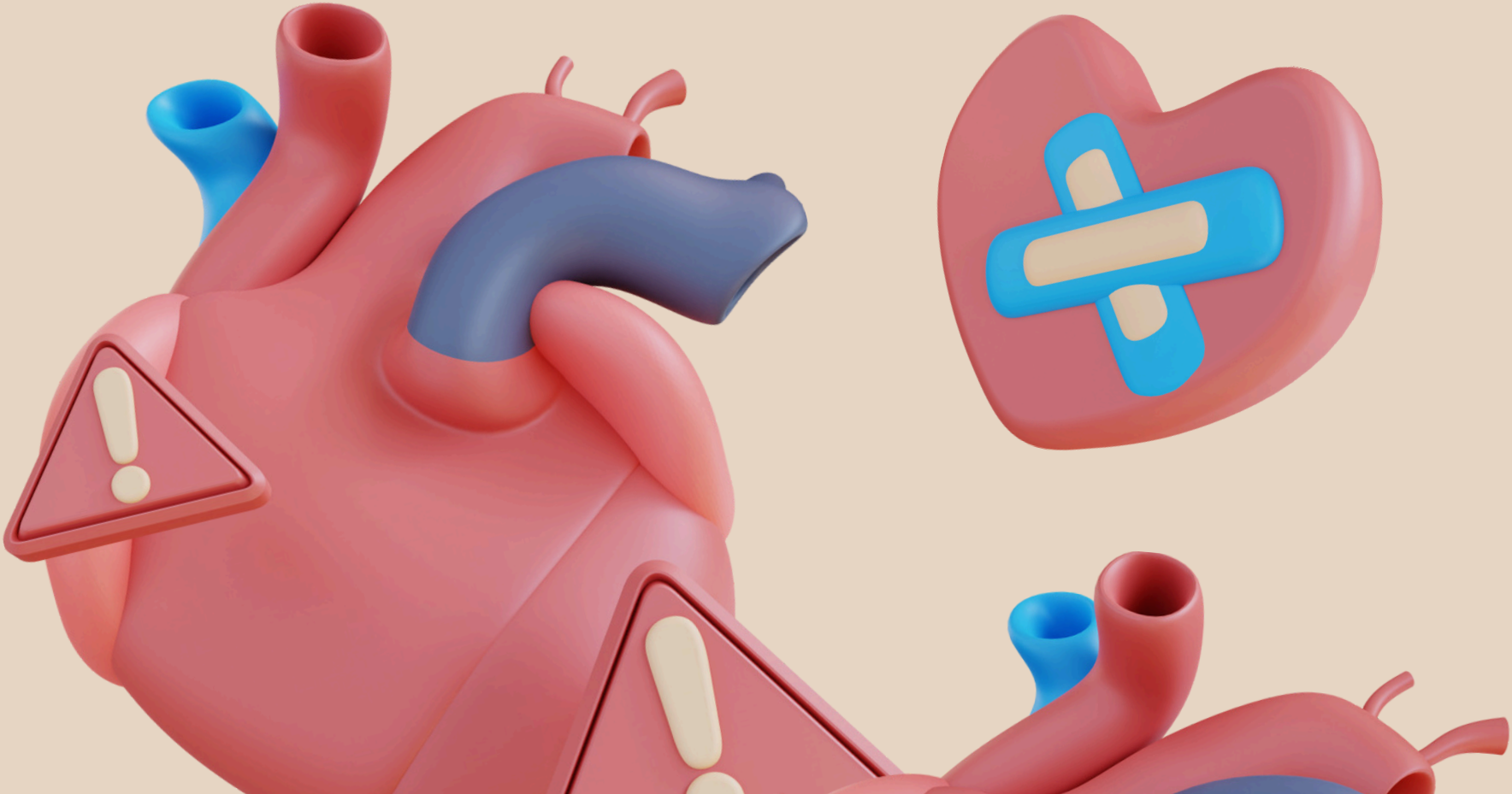
beta[6] (chain3)



beta[7] (chain3)

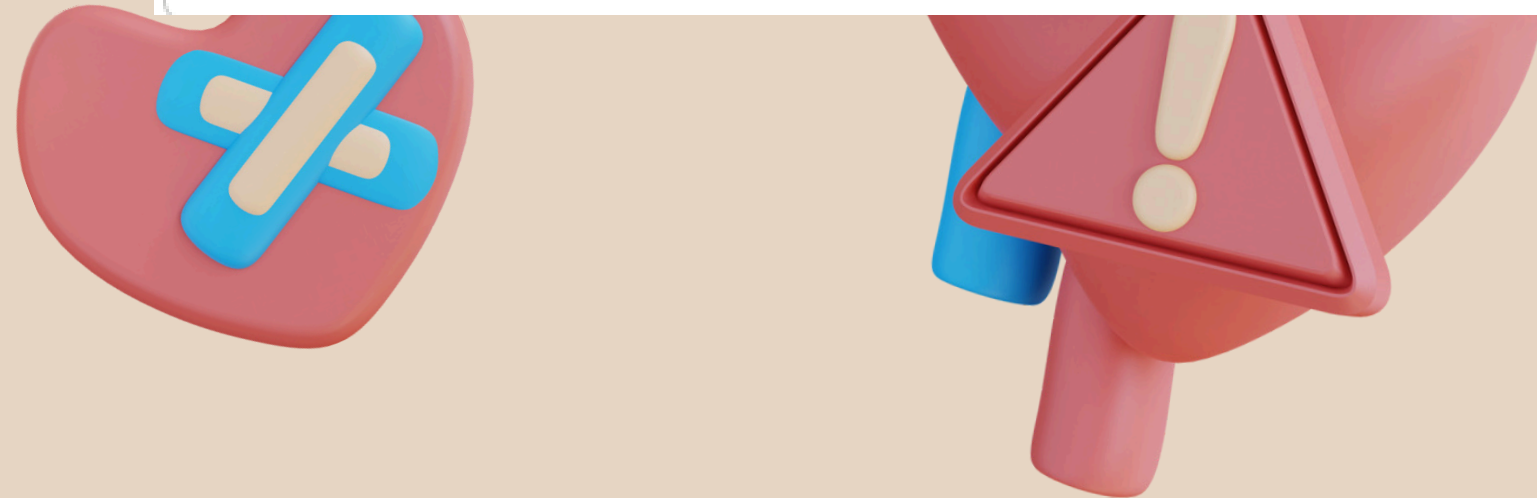


```
## [[3]]
##
## Fraction in 1st window = 0.1
## Fraction in 2nd window = 0.5
##
##      alpha  beta[1]  beta[2]  beta[3]  beta[4]  beta[5]  beta[6]  beta[7]
## -0.19133  1.86314 -1.74706  0.07082 -2.09518  0.09799 -1.51199  0.18409
##  beta[8]  beta[9] beta[10]
##  0.81178 -1.44691  0.92011
```

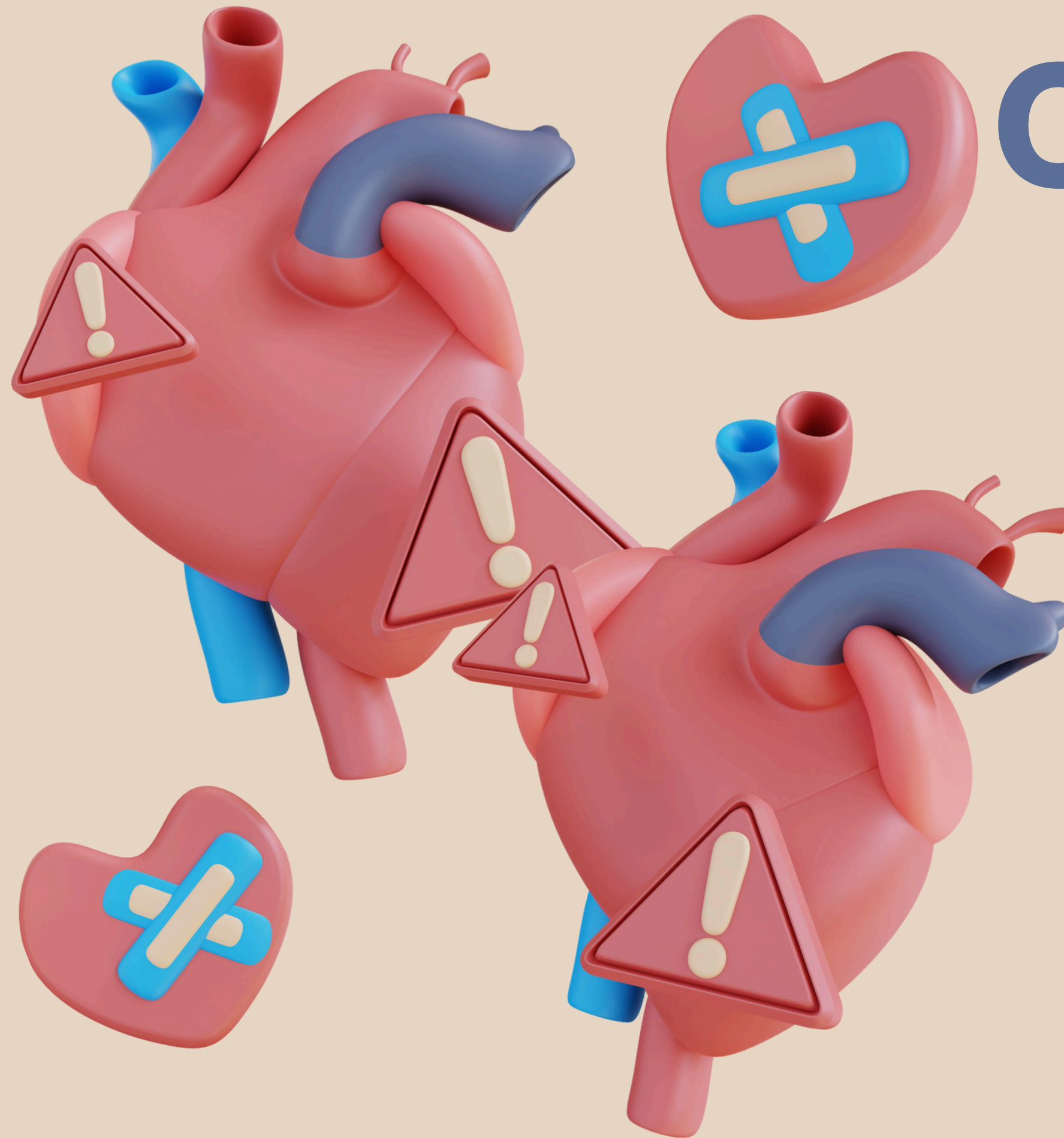



EFFECTIVE SAMPLE SIZE

```
##      alpha  beta[1]  beta[2]  beta[3]  beta[4]  beta[5]  beta[6]  beta[7]
## 4369.365 3428.797 4613.669 3642.618 3426.386 3961.433 3152.011 3680.496
##  beta[8]  beta[9] beta[10]
## 3951.941 4500.000 3285.077
```



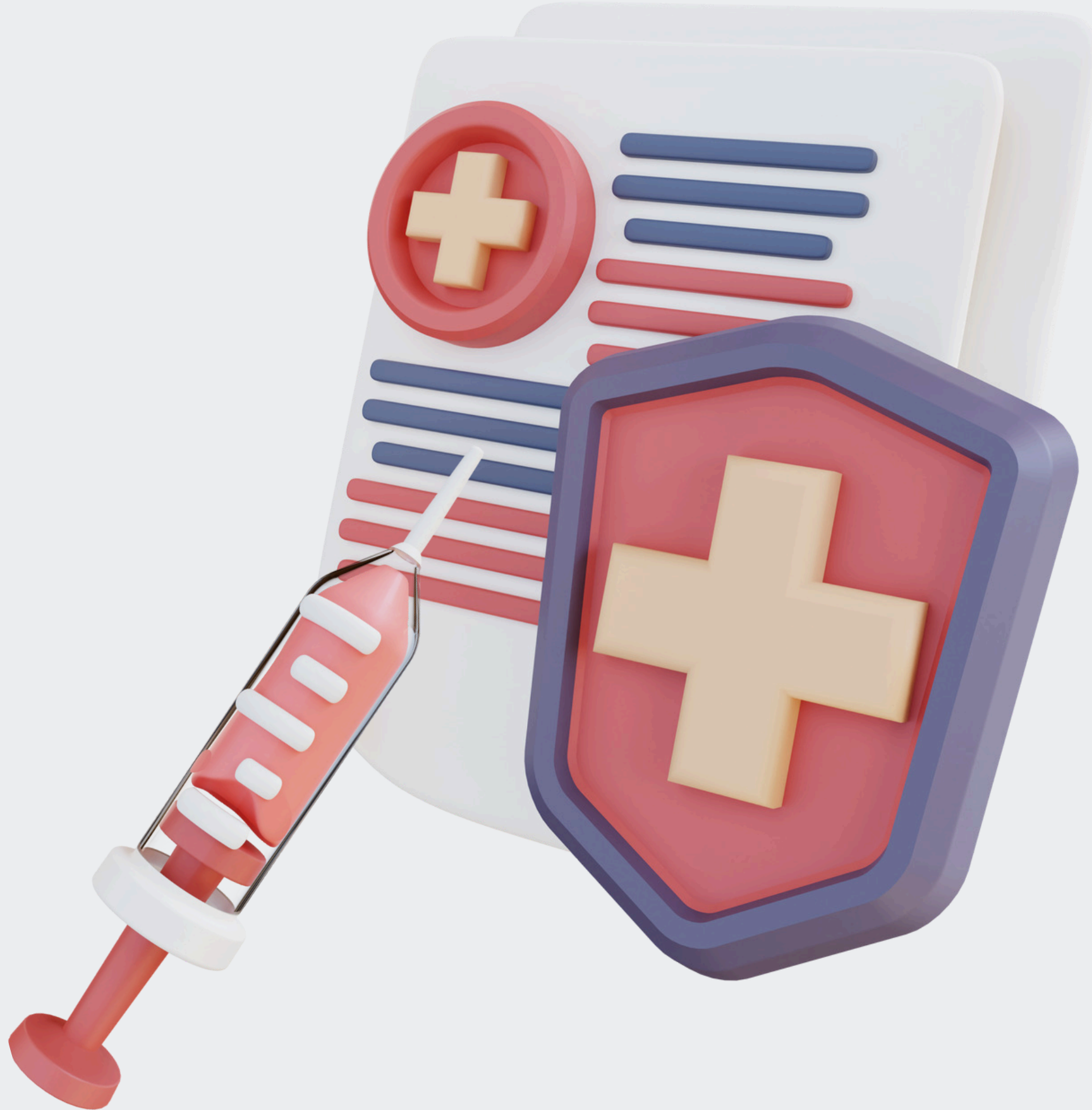
Indicates the “effective” number of independent samples from the posterior distribution.



CONCLUSION

Overall, the Bayesian Logistic Regression provides an effective, interpretable, and robust method for breast cancer malignancy prediction.

Compared to traditional logistic regression, the Bayesian approach offers advantages by delivering full posterior distributions, credible intervals, and principled uncertainty quantification, rather than relying solely on point estimates and asymptotic assumptions.



**THANK
YOU!**